4.1 Types of software

Computers need **software** to allow the user to do something useful. You will use software each time you use a computer, whether this is a PC, a laptop, tablet or mobile phone. Software is a series of instructions that tell the computer what to do, created using programming languages..

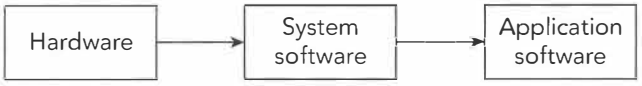


Figure 4.3: System software manages the running of hardware; application software runs on top of system software

Software can be put into different categories, depending on their purpose and use (Figure 4.3 shows the link between hardware and software).

System software manages the running of the hardware and other software, and allows communication between these components. These are tasks that help the computer perform its functions and improve or maintain its efficiency. Two types of system software are the **operating system (OS)** and **utility programs**.

Examples of utility programs are not required but useful to know. These are programs that maintain the computer system, for example:

- System clean-up-searches for and removes un-used programs and data.
- Defragmentation-rearranges files so that they are located together (contiguously) and that the free spaces is located together.

Application software allows the user to perform a task – it provides the services for the user, for example, write a letter, view a website, etc.

Application software have generic names, as well as the brand names. It is important that you know the generic names, some of the common ones are:

- A **word processor** allows the user to create written documents.
- A **spreadsheet** allows the user to perform calculations on data.
- A **database** allows the user to store and search a set of data.
- A **web browser** allows the user to view and navigate websites.

KEY WORDS

software: a series of instructions written in a programming language that performs a function.

system software: a type of software that manages the hardware and software in a computer.

operating system (OS): a program that manages the computer, allows the user to communicate with the computer, and allows software to be run.

utility programs: system software that performs house-keeping activities.

application software: software that allows the user to perform a useful task.

word processor: a type of application software that allows the user to create text-based documents.

spreadsheet: a type of application software that performs calculations on data.

database: an example of application software to store and manipulate data.

web browser: a piece of software that retrieves and displays web pages.

COMPUTER SCIENCE IN CONTEXT

PCs, laptops, tablets and smartphones all have application software. The brand names may be different, but the fundamental purposes are the same. For example, you can open a word processor on a PC, a laptop, a tablet and even a smartphone.

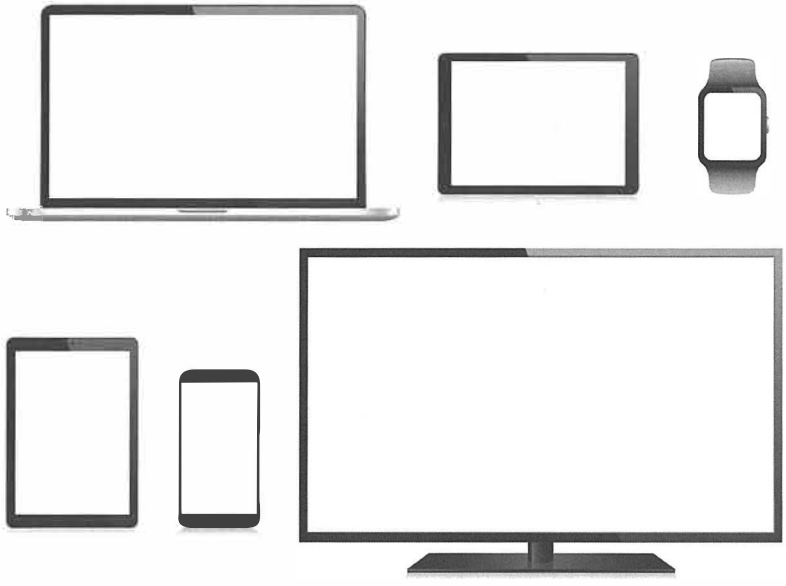


Figure 4.4: Digital devices come in many different forms

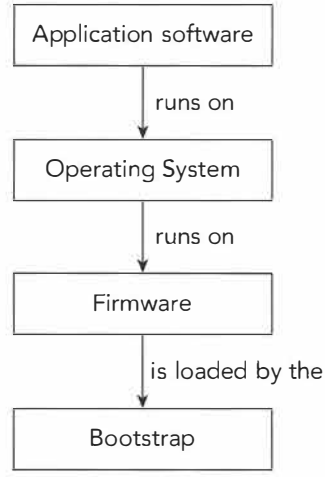


Figure 4.5: The software on a computer.

The application software cannot run on a computer without an operating system. The operating system cannot run unless there instructions given to it, these are provided by the **firmware**. The firmware cannot run unless it is told to run, this is done using the **bootstrap** program, which is loaded when the computer first turns on. This process can be represented as shown in Figure 4.5 on the right.

ACTIVITY 4.1

Log onto a computer, tablet or mobile phone. Make a list of the different types of application software that are installed. Remember not to use brand names.

KEY WORDS

firmware: instructions that are stored in the ROM and are loaded when the computer starts.

bootstrap: the first commands loaded when a computer starts, it checks hardware and loads the firmware.

4.2 Operating system

An operating system (OS) is an example of system software, you may be familiar with brands such as Windows, Linux and macOS for PCs and laptops. Other computers also have operating systems, for example, a mobile phone may have an Android OS, or iOS. The overall purpose of these operating systems is to allow you, the user, to interact with the hardware.

The OS has many different roles that are described in the following sections.

Providing an interface

The OS allows the user to enter data, and outputs data to the user, all through the **interface**. There are different types of interface:

- **Graphical User Interface (GUI)** has windows, icons, menus and pointers (WIMP). It is what you will be familiar with from PCs, mobile phones, etc. This is most useful for novices because it is often intuitive. An example is Windows.
- A **command line interface** requires the user to enter text as commands. This is most useful for expert users because the commands entered must be exact. An example is Linux.
- A **natural language interface** allows the user to enter any text commands, or speak commands. The OS then analyses what has been entered and performs the action.

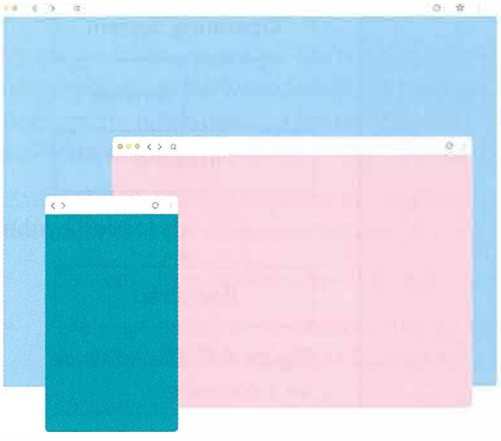


Figure 4.6: A Graphical User Interface

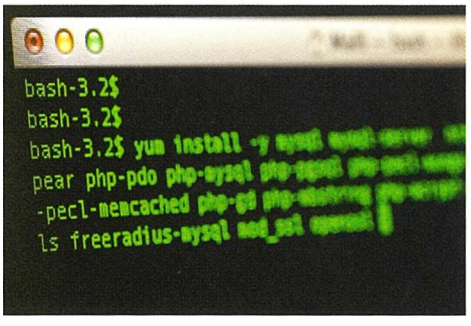


Figure 4.7: An image of a command prompt

KEY WORDS

interface: the method by which a user communicates with a computer.

Graphical User Interface (GUI): a type of operating system that includes windows, icons, menus and pointers.

command line interface: a type of operating system where the user types commands.

natural language interface: a type of operating system that allows the user to type or speak commands.

COMPUTER SCIENCE IN CONTEXT

Have you every spoken a command to a computer? For example, Amazon's Alexa, Microsoft's Cortana or Apple's Siri? These are examples of natural language interfaces. You do not have to speak specific commands, the programs analyse the words you have said and decide on a response. There is more about how this is done in Chapter 6.

Managing files

A user will use application software to create files. The OS allows the creation of directories that the files can be stored within. It also allows the user to move files, copy files, delete files, name and rename files, plus many other actions such as sorting by date.

Managing peripherals and drivers

A computer cannot be used without input and output devices. These are called **peripherals**. Anyone can build a peripheral, and it will be programmed by the person

KEY WORD

peripheral: a hardware device, used to input, store or output data from a computer, that is not directly part of the computer itself.

who made it – not the developer of the OS. This means that it may not be compatible (for example, it may use a different interpretation of binary). A **driver** is software that translates data from the computer to the peripheral, and vice-versa. The OS allows the installation of these drivers and the sending of data to and from it.

KEY WORD

driver: a program that controls a device, for example, a printer or a keyboard.

Managing memory

A computer uses memory to store data. This can be using secondary or primary memory.

The OS manages the movement of data to and from hardware, primarily RAM. It also checks the processes being carried out and makes sure that they have enough memory allocated to perform their function. The OS will also make sure that two processes do not try and access the same memory location.

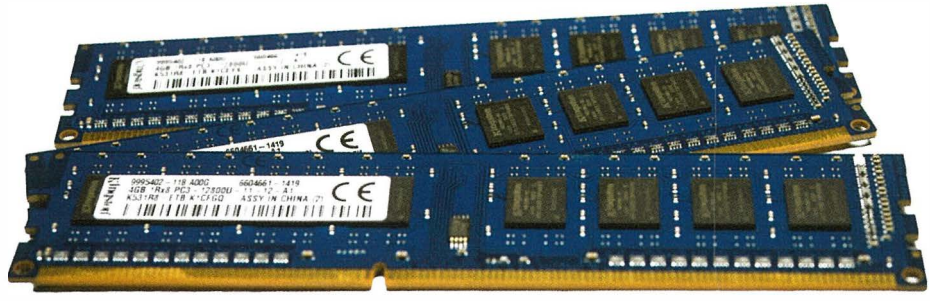


Figure 4.8: RAM memory modules

Managing multitasking

A single processor can only ever execute one instruction at a time. The processor does this so fast that it appears to be doing several tasks at once. For example, you might be listening to music from a computer, typing a document into a word processor, and the computer is outputting what you are doing. The OS allows this to happen by deciding which processes should be executed next and how long they can spend being processed before switching to another process. The OS does this using **interrupts**.

KEY WORD

interrupt: a signal sent to the processor to tell it that its attention is required.

Managing interrupts

An interrupt is a signal sent to the processor to tell it that something needs its attention. The OS decides which interrupt to run and then performs the actions required. There is more about interrupts in Section 4.3.

Providing a platform for running applications

Application software needs to run on a computer. The software could be written by anyone using any language. The OS allows the application software to run on the computer by fetching instructions from it and executing them.

Managing user accounts

Most computers need a user to set up an account. This can include preferences, as well as a username and password (text and/or biometric). The OS allows a user to set-up an account, keeps the data separate for multiple accounts, and restricts the access by using the password.

4.3 Interrupts

An interrupt is a signal to the processor to tell it that something needs its attention. An interrupt can be software or hardware based. Table 4.1 has several examples of each.

Software	Hardware
Division by 0.	Data input, e.g. key pressed on a keyboard, mouse click.
Two processes attempting to access the same memory location.	Error from hardware, e.g. printer out of paper.
Program request for input.	Hardware failure.
Output required.	Hard drive signal that it has read data.
Data required from memory.	New hardware device connected.

Table 4.1: Interrupt examples

An interrupt has a priority level. A high-priority interrupt needs the attention quickly, for example, hardware failure. A low-priority interrupt needs the processor but it's not urgent, for example, data input. This is managed by the **interrupt handler (IH)**.

When an interrupt is generated it is put in a queue according to its priority. The sequence of activities is:

- When the processor finishes its current fetch-decode-execute cycle (or before the starting the next FDE cycle), it checks the interrupt queue.
- It checks whether there is an interrupt with a higher priority than the current task it is processing.
- If it is, it:
  - Stores the current process and fetches the interrupt.
  - Checks the source of the interrupt.
  - Calls the relevant **interrupt service routine (ISR)**. This is a sequence of instructions that handle the interrupt.
  - When finished the stored process is returned to memory or another higher-priority interrupt is fetched.
- If it isn't, it runs another FDE cycle.

KEY WORDS

interrupt handler (IH): a program that organises interrupts into an order based upon priorities.

interrupt service routine (ISR): a program that retrieves an interrupt and performs the required action(s).

Questions

1 Give **two** examples of application software.
2 State the purpose of system software.
3 Describe the relationship between application software and the bootstrap.
4 Identify **three** tasks the file management of an Operating System can perform.
5 Give **two** examples of software interrupts, and **two** examples of hardware interrupts.
6 Describe the process that happens when an interrupt is sent to the processor.